Info gain for humidy :-

		Yes	No
high	4	2	2
normal	4	2	2

for yes.

$$-(2/4 \log_2 2/4 + 2/4 \log_2 2/4)$$

$$-(0.5 \log_2 0.5 + 0.5 \log_2 0.5)$$

$$(-0.5 - 0.5)$$

$$1$$

for no :

$$-(2/4 \log_2 2/4 + 2/4 \log_2 2/4)$$

$$-(0.5 \log_2 0.5 + 0.5 \log_2 0.5)$$

$$(-0.5 - 0.5)$$

$$1$$

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$\Rightarrow -(-0.442 - 0.528)$

$\Rightarrow 0.442 + 0.528$

$\Rightarrow 0.97$

for no.

$$-\left(\frac{1}{3}\log_2\frac{1}{3} + \frac{2}{3}\log_2\frac{2}{3}\right)$$

$$-(0.333\log_2 0.333 + 0.666\log_2 0.666)$$

$$-(-0.528 - 0.390)$$

$\Rightarrow \boxed{0.918}$

for weighted

$$\Rightarrow \frac{5}{8}(0.97) + \frac{3}{8}(0.918)$$

$$= \boxed{0.9505}$$

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Test.

	outlook OL	Temperature T	humidity H	windy W	Rain R
1	Sunny	Hot	high	weak	No
2	Sunny	mild	high	strong	No
3	cloudy	hot	high	weak	Yes
4	Rainy	mild	high	weak	yes
5	Rainy	cool	normal	weak	Yes
6	Rainy	cool	normal	strong	no
7	clody	mild	normal	strong	Yes
8	Sunny	Cool	normal	Weak	no

For entropy of rain :-

4 yes , 4 no

$$\Rightarrow -\left(\frac{4}{8}\log_2\frac{4}{8} + \frac{4}{8}\log_2\frac{4}{8}\right)$$

$$\Rightarrow -\left(0.5\log_2(0.5) + (0.5)\log_2(0.5)\right)$$

$$\Rightarrow -\left(-(0.5) - 0.5\right)$$

$$\Rightarrow 0.5 + 0.5$$

$$\Rightarrow \boxed{1}$$ entropy of rain

$\Rightarrow -(-0.390 - 0.528)$

$\Rightarrow 0.390 + 0.528$

$\Rightarrow \boxed{0.918}$

weighted entropy is

$$H(S/\text{outlook}) = \frac{3}{8}(0.000) + \frac{2}{8}(0.000) + \frac{3}{8}(0.918)$$

$$(0.375)(0.918)$$

$\Rightarrow 0.34425$

$$IG(\text{outlook}) = 1 - 0.34425$$

$$IG(\text{outlook}) = \boxed{0.6557}$$

⇒ Info gain for Temperature:

	Total	yes	NO
Hot	2	1	1
mild	3	2	1
cold	3	1	2
Total	8		

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IG(windy) = ~~0-0~~ 1-0.9505

IG(windy) = 0.0495

⇒ entropies

→ Rain 1

⇒ outlook ⇒ 0.6557

→ Temperature ⇒ 0.6015

→ humidity ⇒ 0

⇒ windy ⇒ 0.0495

⇒ The root node is (0.6557) which is outlook

for weighted entropy:

$$\Rightarrow \frac{4}{8}(1) + \frac{4}{8}(1)$$

$$0.5 + 0.5$$

$$\Rightarrow \boxed{1}$$

IG (humidity):

$$\Rightarrow 1 - 1$$

$$\Rightarrow \boxed{0}$$

IG for windy:

		yes	NO
weak	5	3	2
strong	3	1	2

for yes.

$$-\left(\frac{3}{5}\log_2\frac{3}{5} + \frac{2}{5}\log_2\frac{2}{5}\right)$$

$$-\left(0.6\log_2 0.6 + 0.4\log_2 0.4\right)$$

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info gain for outlook.

for Sunny:

$$-\left(\frac{0}{3}\log_2\frac{0}{3} + \frac{3}{3}\log_2\frac{3}{3}\right)$$

(0 → Yes, 3 → No)

$$\Rightarrow 0.000$$

for cloudy:-

$$\Rightarrow -\left(\frac{2}{2}\log_2\frac{2}{2} + \frac{0}{2}\log_2\frac{0}{2}\right)$$

(2 → Yes, 0 → No)

$$\Rightarrow 0.000$$

for rainy:

$$\Rightarrow -\left(\frac{2}{3}\log_2\frac{2}{3} + \frac{1}{3}\log_2\frac{1}{3}\right)$$

(2 → Yes, 1 → No)

$$\Rightarrow -(0.666\log_2 0.666 + 0.333\log_2 0.333)$$

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⇒ for hot)

⇒ $-\left(\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{2}\log_2\frac{1}{2}\right)$

⇒ $-(0.5\log_2 0.5 + .0.5\log_2 0.5)$

⇒ $-(-0.5-0.5) \Rightarrow 0.5+0.5$

⇒ 1

⇒ for mild :-

⇒ $-\left(\frac{2}{3}\log_2\frac{2}{3} + \frac{1}{3}\log_2\frac{1}{3}\right)$

⇒ $-(0.666\log_2 0.666 + 0.333\log_2 0.333)$

$-(-0.390 - 0.528)$

⇒ ⇒ $0.390 + 0.528$

⇒ 0.918

⇒ for cool:

⇒ $-\left(\frac{1}{3}\log_2\frac{1}{3}+\frac{2}{3}\log_2\frac{2}{3}\right)$

⇒ $-\left(0.333\log_2 0.333+0.666\log_2 0.666\right)$

⇒ $-\left(-0.528-0.390\right)$

⇒ $0.528+0.390$

⇒ 0.918

weighted entropy:

$H(S/\text{temperate})=\frac{2}{8}(1)+\frac{3}{8}(0.918)+\frac{3}{8}(0.918)$

⇒ 0.3985

$(IG/\text{Temp})=1-0.3985$

⇒ 0.6015